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AIGC-empowered smart manufacturing: Prospects and challenges

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ABSTRACT

Generative AI (GenAI), the technology behind Artificial Intelligence Generated Content (AIGC), has emerged as a transformative technology in smart manufacturing. However, its full potential and integration within manufacturing processes remain unexplored. This paper presents a comprehensive framework that aligns a GenAI-centered approach with Product Lifecycle Management (PLM), systematically examining the AIGC landscape and its applications across various manufacturing phases. To ensure accuracy and relevance, a human-in-the-loop pipeline is employed to curate and analyze cutting-edge research. Key contributions of this study include: 1) a holistic perspective on AIGC-empowered smart manufacturing, 2) an in-depth analysis of the current technological landscape, and 3) the identification of critical research challenges and future directions. Additionally, the paper considers Industry 5.0 principles, emphasizing human-centricity, sustainability, and resilience. By fostering discussion and collaboration, this review aims to advance innovation and unlock the full potential of AIGC in smart manufacturing.

TABLE: List of acronyms and abbreviations.

Acronym	Definition	Acronym	Definition
AIGC	Artificial Intelligence-Generated Content	MLLM	Multimodal Large Language Model
AM	Additive Manufacturing	NeRF	Neural Radiance Fields
DDPM	Denoising Diffusion Probabilistic Model	PEFT	Parameter Efficient Fine-tuning
EBM	Energy Based Model	PLM	Product Lifecycle Management
GAN	Generative Adversarial Network	RAG	Retrieval Augmented Generation
GenAI	Generative Artificial Intelligence	RLHF	Reinforcement Learning from Human Feedback
LLM	Large Language Model	VAE	Variational Autoencoder

1. Introduction

In recent years, Artificial Intelligence Generated Content (AIGC), exemplified by OpenAI's ChatGPT [1], Sora [2], and DALL-E [3], has brought significant changes from both academia and the manufacturing industry. AIGC generally refers to structured data generated by Generative AI (GenAI) across various modalities, including structural data, natural language text, code, images, videos, audio, time-series data, etc. This versatility of AIGC allows GenAI technologies to be applied across a wide range of fields, including smart manufacturing. As the industry progresses toward Industry 4.0, even Industry 5.0 [4] —a paradigm shift that emphasizes human-centric, sustainable, and resilient manufacturing practices—the role of GenAI is becoming increasingly significant.

While substantial research has been conducted on various facets of GenAI, the manufacturing sector has yet to fully realize its potential. The integration of GenAI technologies, representatively by Large Language Models (LLMs) and other state-of-the-art models, in manufacturing is still in its nascent stages, presenting numerous opportunities and challenges. To address this research gap, this paper employs a GenAI-centered approach to systematically explore AIGC-empowered smart manufacturing (Fig. 1).

Our research curation process begins by generating seed keywords

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0736-5845/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

focused on GenAI and smart manufacturing. These keywords are then expanded using a GenAI model to cover the broad range of relevant terms, including various architectures and manufacturing applications. The expanded list undergoes validation by domain experts to confirm its relevance. Next, the validated keywords are merged with our preset inclusion and exclusion criteria, which are defined across three main categories: Time, Scope, and Credibility (Table 1).

Considering some important GenAI research or industrial examples might not be captured through the academic paper retrieval process, as much of it is published as technical reports (e.g., GPT-4o System Card), or valuable blog posts that may not be indexed by academic search engines. To address this issue, a human-selected list of key works about general and domain-specific GenAI models and technologies is developed, which incorporates insights from confidential model rankings (e.g., Chatbot Arena [5]), community feedback (e.g., HuggingFace, Reddit), real-world performance rankings (e.g., Artificial Analysis [6]), and industrial practices from multiple creditable sources. For non-academic sources, we assess credibility through a more rigorous evaluation of source reputation, authorship expertise, information origin verification to ensure only high-quality information is incorporated into our review.

Through the implemented human-in-the-loop pipeline, the collection of 214 related works offers an accurate, up-to-date perspective on the role of GenAI in manufacturing, backboned by a meticulously curated list of relevant studies and key works. Based on researching the underlying approaches of 83 influential works, this paper will present a landscape of GenAI technologies in Section 2. In Section 3, the curated list of 33 strongly related papers serves as a foundation for analyzing the current state of AIGC in smart manufacturing and exploring its future directions and challenges.

Several previous perspective papers or surveys have examined GenAI applying in manufacturing from a variety of perspectives, across from generally [7–10] and specifically, including additive manufacturing [11], predictive maintenance and real-time monitoring [12], and Industry 4.0 [13]. However, none of these studies provide a holistic examination of AIGC applications across the entire smart manufacturing ecosystem, encompassing both general technologies and specific implementations. As Fig. 1 depicted, this paper aims to fill that gap by offering an in-depth analysis of AIGC within the context of smart manufacturing, focusing on the opportunities, future directions, and unique challenges. In summary, main contributions are as follows:

1) **Holistic Perspective on AIGC-empowered Smart Manufacturing**: To our best knowledge, this paper is the first paper which provides a holistic perspective on the current landscape in AIGC-empowered Smart Manufacturing. As depicted in Fig. 1, it focuses on enabling technologies, recent advancements, prospects, and associated challenges.
2) **Comprehensive Analysis of AIGC Landscape**: This paper provides the first detailed analysis of the current AIGC landscape and applications in the context of smart manufacturing. It systematically reviews key areas essential for GenAI-empowered smart manufacturing, including the whole deploying process of GenAI models.
3) **Identification of Research Directions and Challenges**: The paper outlines critical research directions, such as GenAI agents, edge, and hybrid AI systems, while also addressing challenges like domain-specific requirements, and explainability, paving the way for future advancements in AIGC-empowered smart manufacturing.

Table 1
Inclusion and exclusion criteria evaluating papers across time, scope, and credibility for this survey.

Category	Inclusion / Exclusion Rule (I/E)
Time	I1, Recent Studies: Papers published within the last 10 years (up to 1 March 2025) will be included. I2, Historical but Impactful: Older papers that have significantly influenced current developments or remain foundational in the field of GenAI or smart manufacturing will be included. E1, Outdated Research: Papers older than five years that no longer influence will be excluded.
Scope	I1, Comprehensive: Include papers that provide a broad overview (e.g., review articles or surveys) of GenAI adopted in smart manufacturing (e.g., Additive Manufacturing) will be included. I2, Specialized: Papers focusing on specific use cases (e.g., G-code troubleshooting), applications (e.g., concept design), or enabler technologies (e.g., deployment technologies) will be included. E1, Out-of-scope: Exclude publications unrelated to both GenAI and smart manufacturing.
Credibility	I1, Peer-reviewed: Include peer-reviewed research papers. I2, Impactful Community Contributions: Incorporate insights from widely recognized community contributions and influential industrial practices. E1, Uncredible Publications: Exclude publications without peer review or from uncredible sources.
Others	E1, Non-English Publications: Publications that are not available in English or have insufficient English translations will be excluded. E2, Duplicated Publications: Publications that are duplicated in scope, without any different valuable insights will be excluded.

As shown in Fig. 2, the remainder of this paper is organized as follows. Section 2 provides an in-depth review of the AIGC landscape, which covers GenAI model infrastructure, operations, and a brief overview of generative tasks. Section 3 examines the integration of AIGC with PLM, focusing on generative tasks throughout all phases, including design, manufacturing, marketing & logistics, operation services, and recycling/remanufacturing. Section 4 outlines future research directions, highlighting trends in GenAI infrastructure and methods. Section 5 addresses the research limitations and challenges in AIGC-empowered Smart Manufacturing, including technical as well as Industry 5.0 concerns. Finally, Section 6 concludes with key takeaways.

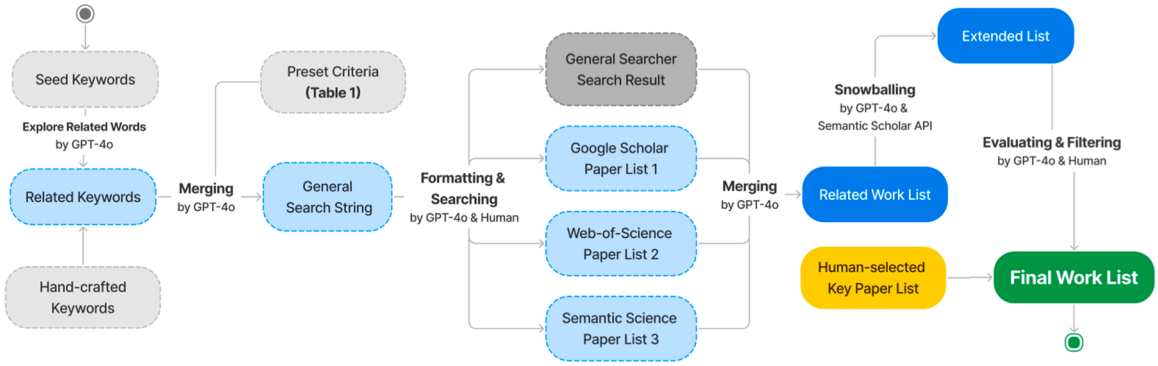


Fig. 1. A GenAI-assisted research and retrieval pipeline.

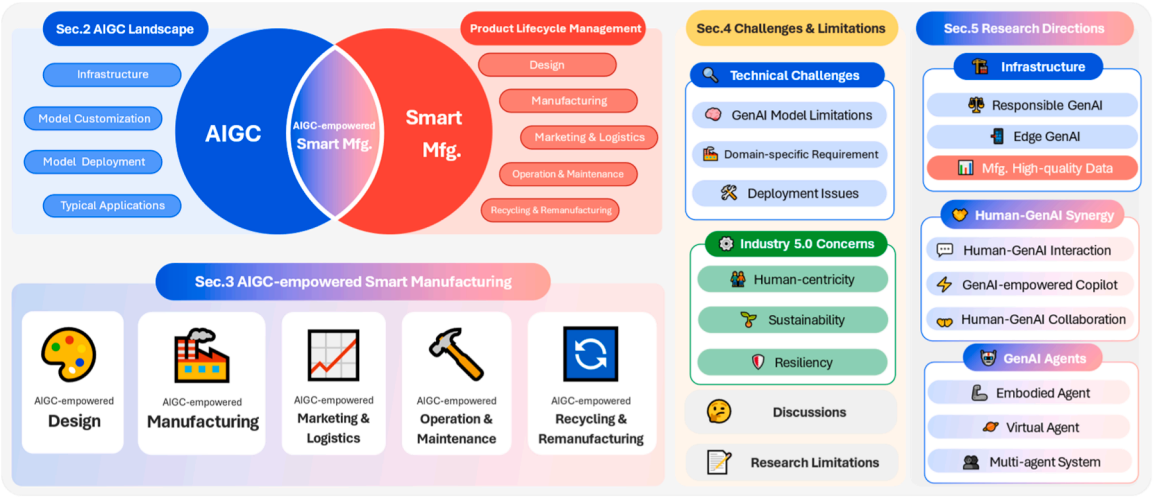


Fig. 2. Overview for this paper.

2. AIGC landscape

2.1. AIGC infrastructure

2.1.1. GenAI model architecture

Traditional AI, which is used solely as a reference point for comparison with GenAI, is based on machine learning and deep learning models that work with structured data. It performs tasks such as predictive analytics, anomaly detection, and optimization functions including predictive maintenance, quality control, and supply chain forecasting. These models use established methods and provide clear interpretability and reliability, which are essential for consistent decision-making. In contrast, GenAI is designed for unstructured data. It supports adaptive and creative tasks like generating designs automatically, interacting with users through natural language, and adjusting manufacturing workflows in real time. Traditional AI achieves high accuracy and efficiency in well-defined tasks, while GenAI offers flexibility and the ability to develop new solutions with improved context understanding. Traditional AI ensures stability by supporting decisions based on structured data, whereas GenAI processes unstructured data to improve human-machine interaction and adaptive learning. This comparison shows that while Traditional AI is a reliable tool for routine tasks, GenAI provides creative potential, even though it needs more resources.

GenAI architectures have evolved significantly over the past decades, especially in recent years as Fig. 3 illustrated, progressing from foundational probabilistic models to sophisticated deep learning frameworks capable of producing highly realistic data across various domains. Before the advent of deep learning, generative models were primarily grounded in statistical methods and probabilistic graphical models [14]. Early models were grounded in statistical methods and probabilistic graphical models (e.g., Hidden Markov Models, Gaussian Mixture Models [15], Boltzmann Machines [16]), which were effective for certain tasks but faced limitations in handling high-dimensional data and capturing complex non-linear relationships.

The introduction of Variational Autoencoders (VAEs) in 2013 marked a significant milestone in deep generative modeling [17]. VAEs integrate neural networks with variational inference to learn continuous latent representations of data. This probabilistic framework allows for the generation of new data samples by sampling from the latent space, making VAEs valuable for tasks such as data generation, semi-supervised learning [18], and anomaly detection [19]. Subsequent works have extended VAEs to handle discrete latent variables [20] and incorporated hierarchical structures [21] to improve representation learning.

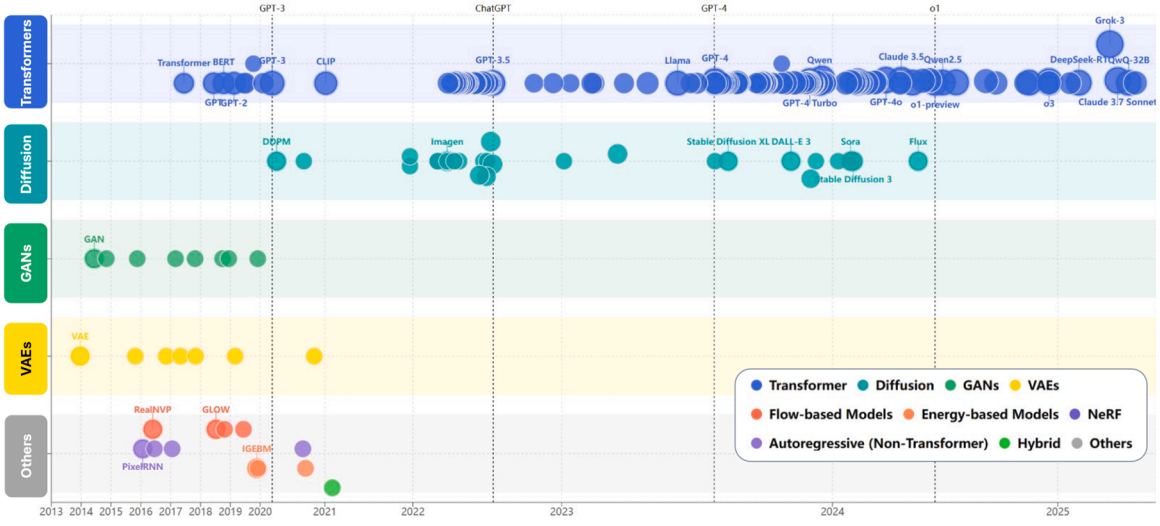


Fig. 3. Overview of 200+ Notable Generative AI models categorized by type and release year (Key models with significant community and industry impact are highlighted).

In 2014, Generative Adversarial Networks (GANs) revolutionized the field by proposing an adversarial framework for training generative models [22]. GANs consist of two neural networks—a generator and a discriminator—that engage in a minimax game. The generator aims to produce data indistinguishable from real data, while the discriminator attempts to differentiate between real and generated samples. This adversarial process propels the generator to produce increasingly realistic outputs. Several variants of GANs have been developed to enhance their performance, such as Deep Convolutional GANs (DCGANs) [23], Conditional GANs (cGANs) [24], and Wasserstein GANs [25], offering improvements by improving image quality, enabling controlled output, and increasing training stability, respectively. GANs have become widely used in applications like image synthesis, style transfer, super-resolution, and data augmentation.

Autoregressive models generate data sequentially, modeling the probability of each element conditioned on previous ones. In 2017, Transformer architecture leveraged self-attention mechanisms to capture long-range dependencies without recurrent structures enabling parallel processing of input data [26]. Transformers and their variations (e.g., BERT [27], Vision Transformers [28]) enable versatile applications across modalities, encompassing both natural language processing (e.g., text summarization, question-answering), as well as computer vision (e.g., image classification, object detection) tasks [29]. Pre-trained models represent a paradigm shift in generative modeling compared to traditional training approaches. Pivoted by Generative Pre-trained Transformers (GPTs) (e.g., GPT-2 [30], GPT-3 [31]), trained on vast text corpora, excel at capturing linguistic patterns and context for generating coherent text. GPT-4 [32] introduced major advancements through its multimodal capabilities, enabling the processing of both text and images, which allows for more complex reasoning and problem-solving, such as interpreting visual data and generating detailed captions. Diffusion models, closely related to score-based models, define a forward process that adds noise to data and a reverse process that learns to denoise and reconstruct the data. Further developments like score-based generative modeling [33] unify diffusion models with score matching, broadening their theoretical foundations. Denoising Diffusion Probabilistic Models (DDPMs) have shown that diffusion models can generate high-quality images comparable to or surpassing GANs [34]. The iterative refinement process offers advantages in training stability and diversity of generated samples. Additionally, applications such as DALL-E [3], Midjourney [35], and Stable Diffusion [36] have showcased the potential of diffusion models in generating detailed and diverse images from textual descriptions.

Beyond categories above, several other generative models have contributed to the field. Flow-based models (e.g., RealNVP [37], Glow [38]) are utilized for precise density estimation and high-fidelity image generation. Neural Radiance Fields (NeRF) based models generate high-quality 3D reconstructions from 2D images by modeling radiance and density volumetrically, particularly useful in 3D scene synthesis and rendering [39]. Energy-Based Models (EBMs) are applied in unsupervised learning and feature extraction tasks, enabling the discovery of complex data representations [40].

In various manufacturing scenarios, each GenAI architecture presents distinct advantages and disadvantages (Table 2). In general, Transformers could aid in generating detailed design specifications or optimizing complex supply chain logistics modeled as sequences, but may not be ideal for visual tasks. GANs shine in visual applications like defect detection and generating synthetic training data for quality control, as well as in novel product design, though their training instability can be a hurdle. VAEs offer a more stable approach for generating design variations and potentially for anomaly detection but might lack the high fidelity needed for critical visual inspections. Diffusion Models promise superior image quality for design and defect identification, but their high computational cost could be limiting. Autoregressive models could be valuable for predicting equipment failures through time-series analysis of sensor data. Flow-based models might excel in modeling intricate manufacturing processes or material properties, offering exact likelihood estimations useful in quality assurance. Finally, Energy-based Models could capture complex relationships in manufacturing data, but their training and inference speed might be too slow for real-time applications. The choice depends on the specific manufacturing challenge, the type and volume of available data, and the criticality of factors like generation quality, training stability, and computational resources.

Table 2
Comparison of generative AI architectures across different facets.

Architecture	Generation Quality	Computational Cost	Inference Speed	Control over Generation	Latent Space	Interpretability	Data Types
Transformers	High, especially for text; improving for images	High for very long sequences	Fast	Good control via prompts and conditioning	Meaningful, but not always explicitly structured	Moderate (attention weights offer some insight)	Primarily text, increasingly images/ audio/video
GANs	Very high realism, especially for images	Very high	Fast	Achievable via conditional GANs, but can be tricky	Implicit, hard to interpret or control	Low	Primarily images/ video, also audio/ other
VAEs	Lower fidelity (blurry) compared to GANs/Diffusion	Moderate	Fast	Control via latent space manipulation, but can be coarse	Structured and continuous, good for interpolation	Moderate (latent space can be analyzed)	Primarily images, also audio/sequential data
Diffusion Models	Very high fidelity and realism, especially for images	Very high	Slow	Good control via conditioning (text, labels)	Often implicit; generation via denoising process	Low	Primarily images, increasing other modalities
Autoregressive	High for sequences (text, audio)	High for long sequences/high-res images	Slow (sequential generation)	Control via conditioning on preceding elements	May or may not have an explicit latent space	High for individual step prediction, but overall model can be complex	Sequential data (text, audio), images (PixelRNN/CNN)
Flow-based Models	High quality, allows exact likelihood estimation	High	Fast	Control via manipulation of the invertible latent space	Well-defined and interpretable due to invertible mapping	High (invertible mapping)	Various, where invertible transformations exist
Energy-based Models	High potential for realism, depends on the energy function	Very high	Slow (sampling required)	Control via conditioning of the energy function	Implicit; energy landscape defines probability	Low	Versatile, including images, audio, text, structured data

2.1.2. Dataset creation & data synthesis

The foundation of any successful GenAI model in industrial applications lies in the quality and relevance of the data used. Developing such models requires a comprehensive data curation strategy that addresses data quality, model robustness, and operational efficiency, tailored to the specific industrial context for continuous monitoring and iterative improvement.

Data curation involves gathering, cleaning, and organizing data from diverse industrial sources such as IoT sensors, machine logs, and enterprise systems. In manufacturing environments, data emerges from a wide array of sources, including IoT sensors, machine logs, ERP systems, and CNC machines. This data may include real-time sensor readings like temperature, vibration, or pressure, as well as operational logs, production schedules, and maintenance records. These data arrive in various formats, such as Time-series data from IoT sensors in CSV, JSON, or proprietary binary formats; Event-based machine logs in structured JSON or XML; and Structured tabular data from ERP systems stored in relational databases like SQL. The data categories include raw numerical time-series data (e.g., temperature or vibration readings), categorical data (e.g., machine states like "on," "off," or "idle"), textual data (e.g., notes from maintenance logs), and derived features (e.g., statistical summaries or frequency domain features from signal processing). Since manufacturing data often exhibits a high velocity, variety, and volume typical of Industrial IoT (IIoT) data, it is crucial to implement standardized data models and interoperability protocols to facilitate integration across heterogeneous systems. Industrial Internet Reference Architectures (IIRA) and standards like OPC UA help maintain consistency and enable the seamless use of data across multiple platforms.

In cases where real-world data is scarce or incomplete, synthetic data augmentation can enrich datasets without introducing bias [41], thereby enhancing model performance. After curating data, meticulous partitioning and thoughtful integration of datasets are essential for effective model training, parameter optimization, and unbiased performance evaluation on unseen data, thereby improving generalizability and robustness. Given that industrial data often contains sensitive or proprietary information, ensuring compliance with data access and copyright regulations is critical. Proper licensing and permissions are essential to protect both data owners and organizations.

2.1.3. GenAI model training

The training of GenAI models relies on several key technologies and approaches, informed by the understanding of scaling laws. Scaling laws describe how model performance improves with increases in model size, data volume, and computational resources, following predictable patterns with diminishing returns at extreme scales [42]. For instance, generative models like GPT-4 [32] have shown that scaling up leads to emergent abilities and better generalization. This principle has driven the development of advanced training technologies, such as large-scale foundation models and multi-modal training techniques, while also highlighting challenges related to efficiency and costs.

Pre-trained Foundation Models. Foundation models are large neural networks trained on extensive and diverse datasets, enabling them suitable for a wide range of applications across various industries [43]. By capitalizing on the insights from scaling laws, these models achieve broad capabilities that allow them to be fine-tuned for specific tasks, eliminating the need for task-specific architectures. Prominent examples, such as Llama [44], BLOOM [45], and GPTs [31,32,46] showcase their broad applicability across a wide range of applications.

Reinforcement Learning for GenAI. In training GenAI models, especially LLMs, the main usages of RL include reward model training and preference fine-tuning. Reinforcement Learning from Human Feedback (RLHF) emerged as a significant advancement in AI training methodologies, notably proposed by OpenAI [47]. This method is used to align complex AI systems with human preferences, enhancing model performance by incorporating qualitative human feedback directly into the training loop. RLHF is widely applied to align with human values and general performance optimization (e.g., InstructGPT [48]). In manufacturing, RLHF can train models to adhere to industry standards and safety protocols by leveraging expert feedback, such as optimizing robotic assembly for safety and efficiency.

Multi-modal Training. Multimodal training involves integrating and processing data from multiple modalities, including text, images, audio, and video—to create models with a more comprehensive understanding of information [49]. This approach, often enabled by scaling to handle diverse data types, supports advanced capabilities like image captioning, text-to-speech synthesis, and video analysis. Techniques like cross-modal attention [50], and shared representation learning [51], are employed to align and integrate features from different modalities effectively. Besides, Vision-Language Models (VLMs) and Vision-Language-Action Models (VLAs) are based on pre-trained backbones that integrate multimodal information. VLMs (e.g., CLIP [52], GPT-4V, Qwen-VL [53]) focus on understanding and generating image-text representations, enabling applications like captioning, visual question answering, and cross-modal retrieval. Building on VLMs, VLAs (e.g., OpenVLA [54], $\pi 0$ [55], MDT [56]) extend this capability by incorporating action representation, introducing additional tokens that encode both the robot's state and its motor commands. This augmentation allows VLAs to model interactions within an environment, making them suitable for robotic control and decision-making tasks.

2.1.4. Industrial GenAI

Industrial GenAI is a specialized branch of GenAI designed for industrial applications. While general-purpose GenAI models are trained on diverse datasets to support tasks such as content creation, conversational AI, and general reasoning, Industrial GenAI is built on models trained with extensive domain knowledge and industry-specific datasets (Table 3). This allows it to generate specialized outputs, including design recommendations, process optimizations, and technical documentation, all tailored to complex industrial workflows. By embedding industrial expertise, these models enhance intelligent decision-making, enable real-time control, and support comprehensive lifecycle management.

As GenAI technology has advanced, domain-specific industrial models have emerged to address the unique challenges of complex production environments. Unlike general-purpose GenAI systems, these models integrate deep domain knowledge with strong reasoning capabilities and workflow-specific logic. They are designed for sectors such as healthcare (e.g., BioGPT, MedPaLM), mining (e.g., Pangu Mining Model), large-scale enterprise applications (e.g., Claude Enterprise), and general industrial use cases (e.g., COSMO-GPT, AInnoGC). Industrial GenAI supports tasks such as anomaly detection in equipment operations, compliance verification in regulated environments, code generation for manufacturing processes, and domain-specific question answering. In regions with advanced computational infrastructure, well-established industrial ecosystems, and a high demand for automation, the development and deployment of these specialized models have accelerated. This combination of factors enhances model accuracy, reliability, and adaptability across diverse production contexts.

Despite these advancements, many existing Industrial GenAI solutions rely on compute-intensive architectures and focus primarily on tasks such as conceptual design or vision-based quality inspection. However, they often struggle with more complex scenarios. For instance, optimizing process parameters in high-end equipment manufacturing remains challenging due to limitations in interpretability, robustness, and practical utility. As Industrial GenAI continues to leverage large-scale training datasets, sophisticated architectures, and modern hardware, its capabilities are expected to expand. Over time, these models will be able to handle more intricate tasks, improve decision-making under uncertainty, and drive the intelligent transformation of manufacturing on a global scale. By refining their underlying technologies and deepening their industry-specific knowledge,

Table 3
Representative Industrial GenAI models, domains, applications, and technical indicators.

Name	Time	Organization	Domain	Industrial Applications	Technical Indicators
Pangu Mining Model	2023	Huawei	Mining	Belt anomaly detection, operational compliance checks, safety monitoring	Anomaly detection accuracy, compliance recognition
BioGPT	2022	Microsoft Research	Healthcare/ Pharma	Drug discovery, healthcare automation	Gene sequence analysis, biological relevance
MedPaLM	2023	Google Research	Healthcare	Medical Q&A, clinical decision support	Diagnostic accuracy, symptom matching
COSMO-GPT	2023	Haier	Machine Vision and Inspection	High-volume visual data analysis, real-time defect recognition, feature extraction	High-resolution feature detection, stable multi-step inspections
Claude Enterprise	2024	Anthropic	Multiple Industry	Engineering, Marketing, Sales, Product Management, Human Resources	Process optimization, Customer segmentation, Conversion rates, Employee metrics
AInnoGC	2023	AInnovation	Multiple Industry	Code generation, multi-stage workflow automation, agent-based reasoning	Reliable code outputs, improved automation of intricate tasks
Lingyang Industrial Model	2024	iFlyTek	Multiple Industry	Industrial content generation, domain-specific Q&A, logical inference, code generation, multimodality	Anomaly detection, predictive maintenance, process optimization, decision-making efficiency, accuracy in domain-specific tasks

Industrial GenAI will move beyond basic design and inspection roles to contribute significantly to process optimization, predictive maintenance, and lifecycle management across the industrial value chain.

2.2. GenAI model customization & deployment

2.2.1. Customization & alignment methods

Customizing generative models for specific applications involves selecting appropriate methods that balance performance, data requirements, deployment costs, and alignment with specific operational goals. Generally, customization methods could be categorized by training status, pre-training, training, and post-training (Table 4).

Train Generative Models from scratch. Building a generative model from the ground up using a vast amount of domain-specific data, significant computational resources, and extensive expertise offers maximum customization and control over the architecture. This allows the models highly tailored to specific industrial tasks, enabling them to meet unique requirements across diverse applications. However, compared to other methods, this approach is extremely costly, time-consuming, and environmentally unfriendly due to the massive carbon footprint associated with the large-scale computations involved [57].

Fine-tuning. Fine-tuning involves adapting a pre-trained model to a specific task or domain by continuing the training process on a smaller, task-specific dataset. This method allows leveraging the general knowledge captured by the pre-trained model while tailoring it to perform optimally on the target application. As a result, fine-tuning is more efficient than training from scratch, as well as leading to greater performance compared to prompt engineering. Fine-tuning techniques are generally categorized into two main approaches: full fine-tuning, and parameter efficient fine-tuning (PEFT), such as Adapter [58] and Low-Rank Adaption (LoRA) [59], which is more commonly used.

Prompt Engineering. Prompt engineering, also known as in-context prompting, involves the strategic design and optimization of prompts used to elicit desired behaviors from LLMs without modifying their parameters. Prompt engineering can significantly influence the quality and relevance of the generated content and is a cost-effective method for guiding model behavior. Recent advancements mainly focus on base prompting (e.g., Few-shot Prompting [60]), models' reasoning enhancement (e.g., Chain-of-Thought [61], Tree-of-Thought [62]), automatic prompt design (e.g., Prefix-Tuning [63]), and other techniques like utilizing external tools (e.g., Program-aided Language Models [64]).

Retrieval-Augmented Generation (RAG). Original RAG enhances the model's outputs by integrating external knowledge sources during inference [65]. To address hallucinations, i.e., when the model generates inaccurate or fabricated information that appears plausible but is not factual, especially in domain-specific tasks, RAG supplies up-to-date information without requiring extensive retraining. Recent works mainly focus on specific improvements based on the naïve RAG, such as incorporating modular components for enhanced adaptability, and retrieval quality improvements with fine-grained indexing and reranking methods [66].

2.2.2. Test & evaluation

A reliable test and evaluation framework is essential for obtaining accurate and unbiased assessments of GenAI systems in industrial settings. In these settings, validating system performance, safety, and alignment with actual operating conditions and business objectives is challenging. Rapid progress in GenAI has made traditional evaluation methods insufficient. This shift has led to the development of advanced methods that incorporate human feedback, dynamic assessments, and real-world performance metrics to ensure that models meet specific industrial needs.

Generative Evaluation & Challenges. Evaluation methods for generative models can be classified into three categories: automated evaluations, human-as-a-judge assessments, and model-as-a-judge approaches [67]. Automated evaluations utilize quantitative metrics such as perplexity, BLEU (Bilingual Evaluation Understudy) [68]. Human-as-a-judge assessments involve experts who provide qualitative judgments on the model's outputs, capturing nuances that automated metrics may miss, such as relevance to industrial processes and compliance with safety standards. Model-as-a-judge approaches employ auxiliary models to evaluate outputs, which can streamline the evaluation process but may introduce their own biases, necessitating careful validation, especially in safety-critical industrial environments. Current challenges include issues with hallucinations, bias, interpretability, and robustness. New evaluation frameworks, datasets, and methods have been introduced to address these issues. Examples include human-in-the-loop assessments (e.g., Arena-hard [69]), hallucination detection (e.g., HaluEval [70]), language understanding (e.g., MMLU-Pro [71]) and alignment (e.g., AlignBench [72]).

Red-teaming. Red-teaming is a proactive strategy that involves deliberately challenging a model with adversarial inputs to uncover vulnerabilities, biases, or unintended behaviors [73]. This approach is based on a threat model that defines key elements such as the system under test, potential adversarial actors, and the tactics, techniques, and procedures (TTPs) they may use, as well as the impact of any identified weaknesses [74]. In industrial settings, red-teaming is applied not only at the model level but also across entire systems. For instance, in a predictive maintenance system for industrial machinery, adversarial scenarios may involve injecting anomalous sensor readings that simulate rare but critical failure modes. Balancing adversarial robustness, defensive adaptability, and model usability poses a fundamental challenge in red-teaming, as excessive blocking can hinder legitimate queries, while insufficient safeguards leave models vulnerable to exploitation. Recent methods include automated attack generation by

Table 4
Comparison of GenAI model customization & alignment methods for industrial applications.

Phase	Techniques	Description	Data Requirement	Deploy Cost	Pros.	Cons.	Ref.
Pre-training	Train from scratch	Training a new model entirely using domain-specific data.	Large amounts of labeled data	Extremely High	Tailored performance; full control over model architecture.	Time-consuming; resource-intensive; requires expertise.	[57]
Training	RLHF	Refining model behavior using reinforcement learning guided by human feedback.	Data on human preferences and feedback	High	Improved alignment with human values; enhanced safety.	Expensive; complex implementation; requires human labor.	[47]
	Fine-tuning	Adjusting a pre-trained model with additional domain-specific data.	Moderate amounts of labeled data	Moderate	Efficient; leverages existing models; faster deployment.	May inherit biases; limited by base model architecture.	[58,59]
Post-training	Prompt Engineering	Crafting inputs to elicit desired outputs without changing model parameters.	Minimal data	Low	Quick implementation; no retraining needed.	Limited control; may not generalize; less robust.	[60–64]
	RAG	Augmenting model outputs by retrieving information from external data sources.	Minimal additional data; access to databases	Low to Moderate	Reduces hallucinations; up-to-date information.	Dependency on data sources; potential latency issues.	[65,66]

Reinforcement Learning (e.g., LLMStinger [75]), Prompt Engineering (e.g., PreciseDibias [76]), and multimodal jailbreaking (e.g., Arondight [77]). Defense strategies focus on modifying the decoding process or adjusting prompt inputs to counter jailbreak attacks while addressing issues of false-refusal and over-refusal. False-refusal occurs when a model mistakenly declines a safe query because it resembles harmful prompts, whereas over-refusal refers to the excessive rejection of benign queries, reducing responsiveness. Mitigation techniques such as Single Vector Ablation remove activation patterns that cause unnecessary refusals [78]. OR-Bench provides a benchmark dataset for systematic evaluation of over-refusal rates [79], and predictive modeling of refusal behavior helps refine safety constraints by analyzing input patterns and adjusting response mechanisms accordingly [80].

Industrial Evaluation Framework. Developing robust evaluation datasets and frameworks for GenAI models in industrial settings is essential for achieving both technical quality and alignment with business objectives. Task-specific metrics—such as improvements in operational efficiency, reduction in defect rates, or minimized downtime—ensure that evaluations reflect the true business impact. Increasingly, real-world performance benchmarks are being incorporated into these frameworks. Several case studies from sectors such as manufacturing and logistics have demonstrated that linking evaluation metrics to key performance indicators (KPIs) and return on investment (ROI) can lead to measurable improvements in operational performance. These real-world examples underline the practical benefits of adopting evaluation frameworks that integrate technical rigor with business performance measures.

2.2.3. Inference

Efficient Inference. Inference is a technical and resource-intensive process that involves converting trained models into actionable insights or outputs, requiring substantial computational resources. Optimizing inference not only enhances performance but also reduces costs and latency, which is crucial for real-time or large-scale applications. Achieving efficient inference requires addressing key factors related to computation, memory, and resource management. Optimizing models for efficient inference is essential to meet the performance requirements of real-time industrial applications. For efficient inference, the industry would consider saving inference time, memory, or costs. To achieve the objectives, techniques such as model quantization [81], model pruning [82], knowledge distillation [83], and mixture-of-experts (MoEs) [84] can significantly reduce latency and computational overhead without substantially sacrificing accuracy.

Test-time Compute (TTC). TTC refers to the computational resources utilized during inference, the phase where it generates responses or performs tasks post-training. Traditionally, GenAI systems prioritized minimal inference time for speed, but scaling TTC shifts this focus by increasing computational power during inference to enhance reasoning, especially for complex problems. This approach, demonstrated by models such as DeepSeek-r1 (notably R1-zero) and OpenAI's o1 and o3 series, has renewed interest in RL as a potential alternative training paradigm, particularly when the system achieves an "aha-moment" in problem solving. This method employs slow, deliberate, and logical reasoning to boost accuracy, though it raises compute costs. For example, DeepSeek-R1 leverages RL via Group Relative Policy Optimization (GRPO) to encourage step-by-step reasoning, markedly improving performance on benchmarks like AIME [85]. In multimodal contexts, techniques such as fine-tuning long-form text examples enhance reasoning across diverse data types, while Collective Monte Carlo Tree Search fosters collaborative learning for superior multimodal reasoning [86]. These approaches enable GenAI to mimic human-like, in-depth problem-solving, significantly elevating performance on challenging tasks. However, scaling TTC increases computational expense and latency, necessitating careful optimization. This paradigm promises to advance AI reasoning capabilities substantially. In manufacturing, increased test-time computing can be applied to optimize automated

quality inspection by processing sensor data and images with detailed reasoning, which improves defect detection and system response. In industrial settings, this approach supports simulations of production processes and machinery behavior, aiding in predictive maintenance and more flexible production scheduling.

2.2.4. Deployment

Deploying GenAI models requires careful consideration of various factors, including scalability, reliability, and cost efficiency. The deployment strategies used for GenAI models are critical to ensuring that the models operate effectively in production environments while maintaining performance across different use cases.

Cloud and Edge Deployment. Deployment strategies can involve cloud computing, edge computing, or a hybrid approach. Cloud deployment offers scalability and centralized management, making it suitable for applications with high computational demands and variable workloads. Edge deployment brings computation closer to data sources, reducing latency and enhancing data privacy, which is critical in environments where real-time processing and data security are paramount [87].

Open vs closed models' deployment. Open models (e.g., Llama [44], DeepSeek R1 [85], Qwen [88]) are those whose weights are available to the public, allowing anyone with the right hardware to run, modify, or fine-tune them, while closed models (e.g., OpenAI o1, o3, Claude 3.7 Sonnet) are provided only via APIs or managed services, keeping their internal parameters hidden. Open models are often deployed by communities or research groups on local or cloud systems, which encourages examination and modification, whereas closed models are usually hosted on centralized servers to ensure stable and secure performance. The trade-off lies between transparency and flexibility on one side and controlled access and reliability on the other. Closed models often achieve state-of-the-art performance by leveraging large training resources. Meanwhile, open models continue to improve, sometimes matching closed models while operating at lower compute costs, though often with a performance lag.

Monitoring and Maintenance. Monitoring and Maintenance are also vital parts of the manufacturing industry. Continuous monitoring of deployed models is vital to detect performance degradation, concept drift, or emerging issues. Implementing monitoring tools that track key performance indicators enables timely interventions. Maintenance activities may include updating the model with new data, retraining to adapt to changing conditions, and ensuring compliance with evolving regulations and standards. Continuous feedback and model updating are integral to maintaining the relevance and effectiveness of the AI system. Implementing mechanisms for ongoing monitoring and gathering feedback from the deployment environment allows organizations to identify areas for improvement promptly. Regular updates to the dataset and the model ensure they reflect changes in industrial processes and remain aligned with evolving business objectives and compliance requirements.

Integration with Existed Systems. The integration of GenAI into various types of systems, including intelligent manufacturing systems, is accelerating, enabling broader adoption and more seamless deployment of models across tasks and domains. Initiatives are enhancing interoperability and facilitating the development of sophisticated GenAI-driven solutions. Early explorations in the manufacturing industry have also gained momentum, for example, Industrial-GPT [89] is presented as a concept for integrating GenAI into manufacturing, combining pre-training, fine-tuning, and reinforcement learning on industrial data, along with a Model-as-a-Service (MaaS) approach for flexible, scenario-specific GenAI solutions in smart manufacturing. Further research will continue to refine and expand the integration of GenAI models such as LLMs, diffusion models, as well as Flow-based models into intelligent manufacturing ecosystems, with a focus on improving efficiency, scalability, and customization.

2.3. Typical GenAI applications

The integration of GenAI across different perceptual modalities—language, visual, auditory, action, and multimodal—enables comprehensive data processing and intelligent decision-making (Table 5). This section explores the typical GenAI applications within each data modality, highlighting representative models and architectures that drive advancements in manufacturing.

2.3.1. Language & auditory tasks

Language tasks involve the processing and generation of textual and speech data to facilitate communication, documentation, and automation. Textual data types, such as maintenance logs and production reports, are essential for monitoring operations and diagnosing issues. Models like BERT [90] have significantly improved natural language understanding, facilitating tasks such as sentiment analysis, question answering, and language translation. In the realm of language generation, InstructGPT [91] represents advancements in producing coherent and contextually relevant text based on user instructions. In speech-related tasks, models like Whisper [92] enable high-accuracy automatic speech recognition across multiple languages, while Voicebox [93] advances speech synthesis, allowing for natural and expressive voice generation. Auditory tasks, involving audio perception and generation—well-supported by models like AudioLM [94]—are essential for condition monitoring and predictive maintenance, as operational noises and anomalies in machine sounds provide critical diagnostic information.

2.3.2. Visual tasks

Visual tasks encompass processing and generating images, videos, and 3D content (e.g., CAD models), enabling applications such as image synthesis, object recognition, and scene understanding. Models like Midjourney [35], and Stable Diffusion [36] have revolutionized image generation by producing high-quality images from textual descriptions using diffusion processes. In video analysis and generation, deep learning models facilitate tasks like video summarization, action recognition, and video synthesis. For 3D content creation, methods like DreamFusion [96] and Magic3D [97] allow the generation of 3D models from text prompts, advancing capabilities in multiple domains including virtual reality and design.

2.3.3. Action tasks

Action tasks pertain to the generation and analysis of sequences that represent motions or operations, such as code generation, time-series

Table 5
Categorization of commonly used manufacturing data by perceptual modalities.

Modality	Data Type	Examples	Representative Models
Language	Text	Maintenance logs, production reports	BERT [90], InstructGPT [91]
	Speech	Voice commands	Whisper [92], Voicebox [93]
Auditory	Audio	Machine sounds	AudioLM [94]
Visual	Image	Product designs, defect images	Midjourney [35], Stable Diffusion [36]
	Video	Assembly line footage, robotic operations	Sora [2], Kling [95]
	3D Content	CAD models, virtual prototypes	DreamFusion [96], Magic3D [97]
Action	Code	CNC codes, robotic programming	Codex [98], Deepseek-coder [99]
	Time-series	Sensor data (e.g., vibration, temperature)	DB-CGAN [100]
	Robot Trajectories	Robotic movements, operational instructions	UniPi [101], RT-X [102]
Multimodal	Multimodal	Annotated images, synchronized sensor data	GPT-4o [103], Gemini [104]

prediction, and robotic trajectories. Models like Codex [98], and DeepSeek-coder [99] have transformed code generation by translating natural language descriptions into executable code, enhancing programming efficiency and accessibility. For time-series data, models like DB-CGAN [100] analyze sensor readings (e.g., vibration, temperature) to predict maintenance needs, enhancing system reliability. In robotic trajectory planning, models like UniPi [101] and RT-X [102] can generate precise movement instructions, improving efficiency. Together, these tasks enable smarter, more adaptive manufacturing and robotics.

2.3.4. Multimodal tasks

GenAI is advancing towards models capable of processing and generating multiple forms of data, such as text, images, and audio. Traditionally, models excelled unimodal—for instance, language models proficient in text comprehension but lacking image processing capabilities, and vice versa. Multimodal tasks involve the comprehension and generation of multimodal information and signals, such as language, image, and sound (Fig. 4).

Multimodal Models (MMs) like GPT-4o [103] and Gemini [104] represent significant advancements in this area, capable of understanding and generating responses that combine textual and visual information. MMs can process image inputs and generate text outputs that describe or analyze the images, enabling applications in image captioning, visual question answering, and cross-modal retrieval. Similarly, models like Gemini, announced by Google, aim to unify multimodal capabilities with advanced reasoning, allowing for more comprehensive AI systems that can interact with the world in a more human-like manner. As the world is replete with multimodal information and signals, perceiving and interacting with the world might be the key to achieving human-level AGI.

Recent developments in multimodal models enhance performance in complex, context-rich environments by integrating diverse data types. Notably, multimodal models like PaLM-E [105] have demonstrated the ability to handle multiple modalities, merging them into cohesive outputs. These advancements enable more versatile AI systems that can understand and generate content across mediums, opening new possibilities in areas like creative applications, robotics, and virtual assistants.

3. AIGC-empowered smart manufacturing through the lens of PLM

The advent of AIGC has ushered in a new era for smart manufacturing, leveraging GenAI to revolutionize various facets of the manufacturing process. PLM serves as a pivotal framework within this context, orchestrating the various stages of a product's existence from inception to end-of-life. As Fig. 5 shows, this section explores how GenAI empowers smart manufacturing across the whole PLM phases, including design, manufacturing, logistics, operation, and recycling and remanufacturing.

3.1. Artificial intelligence generated design

In Product Lifecycle Management (PLM), the design phase transforms product concepts into detailed specifications, optimizing functionality, structure, and performance while meeting user needs and manufacturing constraints. Leveraging GenAI in the design process can reduce production costs, accelerate innovation cycles, enable large-scale customization, and decrease emissions over the lifespan of each product. Generative design enhances the design process by generating optimized solutions, boosting creativity, and automating complex decision-making based on user-defined goals and constraints.

Concept Generation. Concept Generation in the context of smart manufacturing involves developing innovative ideas to enhance production using advanced technologies, while requirement capturing ensures that these ideas are aligned with stakeholder needs and operational goals for successful implementation. GenAI enhances concept generation and requirement capturing in smart manufacturing by efficiently processing data, generating innovative solutions, and accurately defining requirements through predictive insights, with research demonstrating its role in stimulating human innovations [106], concept generation [107], generating synthetic shapes [108–110], and design requirements elicitation [111]. A representative real-world example is athletic apparel manufacturer Under Armour, which integrates GenAI into its concept development and refinement process, making it the first to offer a commercially available 3D-printed performance trainer.

Topology Optimization (TO). TO is a computational design strategy used in manufacturing to determine the optimal distribution of material within a given design space, based on predefined performance criteria such as strength or stiffness. In Additive Manufacturing (AM), TO facilitates the creation of highly efficient, lightweight structures, but must be adapted to incorporate manufacturing constraints to ensure the resulting designs are feasible for production. Recent studies leveraging GenAI models, especially GANs [112,113], in TO accelerate design iterations, reduce computational costs, and discover alternative high-performing structures, enhancing design efficiency and innovation.

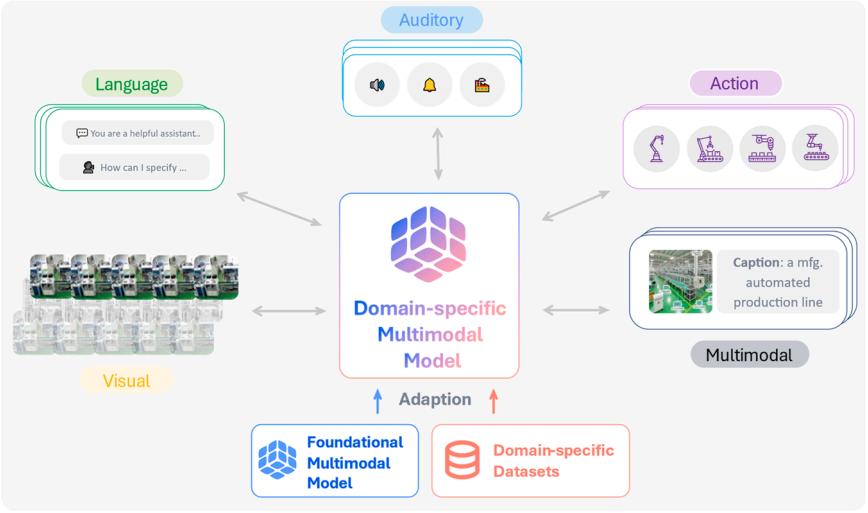


Fig. 4. Multimodal models could function as a universal interface for any-to-any tasks.

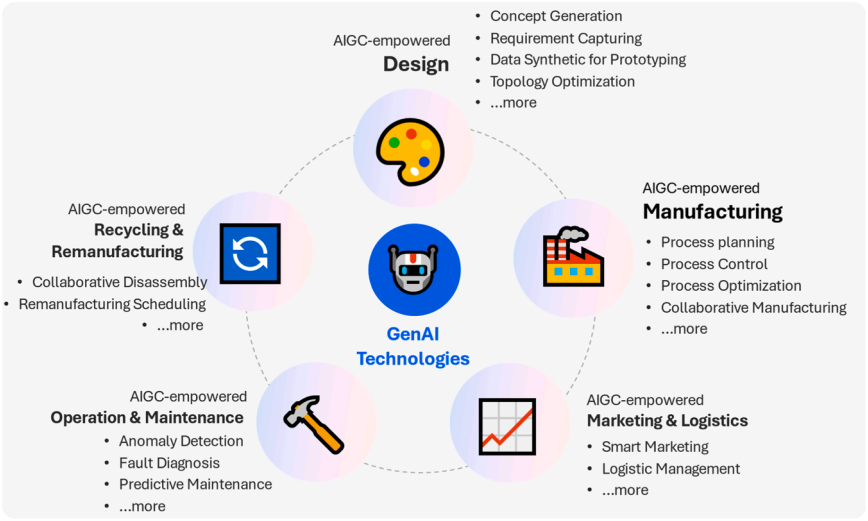


Fig. 5. AIGC empowering the product lifecycle management.

In various industries, enterprises are increasingly adopting generative design to improve to drive innovation, efficiency, and sustainability. Companies such as aerospace giant Airbus and automotive manufacturers like BMW, Ford, and General Motors have leveraged TO in their component design processes to optimize material efficiency. For instance, the BMW i8 Roadster features a roof bracket developed through TO and produced via selective laser melting (SLM), resulting in a component that is 44 % lighter and ten times stiffer than the previous injection-molded version. BMW manufactures approximately 600 of these brackets per build for use in series production vehicles.

3.2. GenAI-empowered manufacturing

Manufacturing in PLM is the pivotal phase where product designs are transformed into physical products. Positioned between design and deployment, it relies on prior development stages and directly affects distribution and service. Manufacturing ensures products are produced according to specifications, on time, and within budget, while also focusing on process efficiency, quality control, and cost management, making it crucial to the overall success of the product lifecycle.

Process planning, control, and optimization. Process planning, control, and optimization ensure efficient production by defining workflows, monitoring quality, and refining processes for better performance. For process planning, current research uses GenAI Agent for process planning of fiber composite structures [114]. Siemens, for instance, a global leader in industrial manufacturing, leverages GenAI to optimize production planning, boosting productivity and reducing costs across its operations. For process control and optimization, several studies utilized GAN-based approaches: Mucllari et al. [115] modeled dynamic welding behaviors for intelligent robotic welding; Zhang et al. [116] combined Long short-term memory (LSTM) and GANs to predict melt pool morphology; Petrik et al. [117] introduced MeltPoolGAN for real-time melt pool classification and generation in Laser Powder Bed Fusion.

Collaborative Manufacturing. Collaborative manufacturing enables real-time coordination between teams and partners, ensuring smooth and adaptable operations. GenAI significantly enhances perception and language processing, enabling more intuitive and effective interactions in a manufacturing context. Recent research has primarily focused on human-robot collaborative manufacturing, where GenAI-driven robots closely cooperate with humans to accomplish complex manufacturing tasks. Studies indicate that GenAI models can analyze real-time production data and environmental information, providing intelligent decision support and optimizing interaction efficiency [118]. Additionally, LLMs based on visual perception and language interaction can support the autonomous navigation and task execution of Automated Guided Vehicles (AGVs) in complex production environments [119]. GenAI empowerment enables mechanical equipment to understand and respond to human needs during the manufacturing process. By combining GenAI with technologies like vision AI, unstructured human-robot collaboration tasks in smart manufacturing can be effectively handled [120,121]. In enterprise-level production, a notable example is the United States Steel Corporation uses Google's GenAI to guide workers through truck repairs and parts ordering, enhancing maintenance efficiency and accuracy. Besides, Siemens Industrial Copilot, developed in collaboration with Microsoft, enhances collaborative manufacturing by automating repetitive tasks, streamlining code generation, and enabling seamless lifecycle human-machine interaction. Similarly, United States Steel's MineMind, a GenAI assistant built on Google Cloud, aids haul-truck repair. Pilot results indicate approximately 20 % faster work-order completion across 60 trucks at the Minntac and Keetac mines.

3.3. GenAI-empowered marketing & logistics

In PLM, smart marketing leverages data analytics and AI to identify target audiences, optimize product positioning, and enhance customer engagement. Smart logistics uses technologies like IoT and AI to streamline supply chains, reduce costs, and ensure timely delivery. Together, they play a crucial role in the product lifecycle, ensuring efficient product launch and successful distribution to customers.

In marketing, GenAI can predict consumer behavior, optimize product features by analyzing market data, and improve demand forecasting. This approach enables more targeted campaigns and informed product development, helping marketers better meet customer needs and drive business success. For example, Wang et al. [122] proposed a Double-GAN model to alleviate data sparsity in e-commerce platforms by integrating user behavior data from multiple sources, enhancing data representation, and reducing computational complexity for user alignment. Marketers also quickly embraced GenAI trends, with tools such as Typeface Arc, Adobe GenStudio or Omneky streamlining enterprise marketing through automated, personalized, multi-channel storytelling that dynamically adapts to audience insights and campaign performance for maximized engagement.

Jackson et al. [123] and Chen et al. [124] both explored state-of-the-art GenAI applications in inventory and logistics management, offering different but complementary approaches. This streamlines logistics planning and decision-making, improving overall

efficiency. Chen et al. introduced a more technical approach, integrating GANs, deep learning, and IoT to optimize logistics routes and enhance customer-focused marketing strategies. The system reduces transportation costs and creates more dynamic, data-driven logistics processes that can adapt to real-time conditions. Big tech companies like Amazon leverage GenAI to optimize logistics and inventory by using real-time data to adapt routes, cut costs, and improve inventory forecasting accuracy. This integration of GenAI with IoT enhances dynamic, data-driven logistics, providing faster and more efficient services.

3.4. GenAI-empowered operation & maintenance service

In the PLM phases, Operation and Maintenance (O&M) ensure efficient production and minimize equipment downtime through daily tasks and timely interventions, playing a vital role in maximizing productivity, ensuring quality, reducing costs, and improving equipment reliability.

Anomaly detection refers to the process of continuously monitoring equipment or system performance to identify deviations from normal behavior, which signal potential issues or malfunctions. Once the anomaly is detected, fault diagnosis is critical for pinpointing specific issues within industrial systems, ensuring timely interventions. GenAI models improve operational efficiency and equipment reliability by addressing imbalanced data and enhancing anomaly detection and fault diagnosis. Various types of generative models are deployed for limited sample augmentation, imbalanced data synthetics, and sensor signal imputation across different tasks in anomaly detection [125], including defect detection for various domains [126–130], and additive manufacturing [131,132].

Predictive maintenance has advanced significantly with the integration of GenAI, which enhances data generation and model optimization. GenAI-enhanced predictive maintenance improves the accuracy of failure predictions by using advanced AI techniques to simulate equipment behavior and generate synthetic data. This allows predictive models to make reliable forecasts even with limited real-world data, leading to more efficient and proactive maintenance strategies. Recent developments mainly include GAN-based [133–135], and VAE-based [136] methods for Remaining Useful Life (RUL) prediction. In industry, companies like General Electric and Eastman use GenAI for predictive maintenance—enabling them to forecast equipment needs and fluid degradation, reduce downtime, and ultimately enhance operational efficiency and cut costs.

3.5. GenAI-empowered recycling & remanufacturing

In the final stages of the product lifecycle, GenAI supports recycling and remanufacturing efforts. GenAI models can design products with end-of-life considerations in mind, facilitating easier disassembly and material recovery. Recent advancement includes human-robot collaborative disassembly [137], where GenAI-driven robots utilize real-time data and cognitive feedback to optimize disassembly, reducing human workload and improving efficiency. For example, A DCGAN-based data augmentation method [138] is illustrated to enhance MI-EEG signals, facilitating human-robot collaboration in disassembly and contributing to smart manufacturing and sustainability. Besides this, Zheng et al. [139] proposed an adaptive rescheduling strategy for aircraft parts remanufacturing using CGAN for data augmentation and IRF for selecting optimal rescheduling methods, enhancing efficiency under disruptions.

Table 6 summarizes GenAI's role across phases in PLM: aiding the design phase in aspects such as concept and prototype generation, or topology optimization; optimizing manufacturing processes with GenAI-empowered methods; enhancing marketing and logistics with consumer insights and inventory management; supporting operations and maintenance through data augmentation; and advancing recycling and remanufacturing by integrating GenAI methods.

4. Challenges and limitations

Despite the promising advancements in AIGC-empowered smart manufacturing, several challenges and limitations hinder its full realization.

4.1. Technical challenges

4.1.1. Limitations of GenAI models

Interpretability and Reliability. Deep generative models face significant challenges in terms of both interpretability and reliability. For

Table 6
Overview of GenAI applications in PLM phases.

Phases	Topics	Typical Data & Type	GenAI Applications	Ref.
Artificial Intelligence-Generated Design	Concept Generation	Design Documents (Text), Innovation Insights (Text)	Stimulating human innovations, generating ideas, concepts, and solutions	[106,107]
	Prototyping	3D Prototypes, Prototype Images (Image)	Creating synthetic models and prototypes	[108–110]
	Requirement Capturing	Requirements (Text), Speech Inputs (Speech)	Automating requirement elicitation and definition	[111]
	Topology Optimization	Structural Simulations (3D Content)	Generating optimized structural designs, accelerating finding optimal material	[112,113]
GenAI-empowered Manufacturing	Process planning	Workflow Schedule (structural data)	Automating and optimizing process workflows	[114]
	Process control and optimization	Sensor Data (Time-series), Process Control Logs (Text)	Enhancing process control through predictive modeling	[115–117]
	Collaborative Manufacturing	Motion Trajectories (Robot Trajectories)	Facilitating human-robot collaboration	[118–121]
GenAI-empowered Marketing & Logistics	Marketing	Consumer Behavior Logs (Time-series)	Predicting consumer behavior and optimizing strategies	[122]
	Inventory & Logistics	Inventory Data (Time-series)	Optimizing logistics and inventory management	[123,124]
GenAI-empowered Operation & Maintenance Service	Anomaly Detection	Sensor Anomalies (Time-series), System Logs (Text)	Improving detection of system anomalies through data augmentation	[125,131,132]
	Fault Diagnosis	Diagnostic Codes (Code), Fault Logs (Text)	Enhancing fault diagnosis through data augmentation	[126–130]
	Predictive Maintenance	Historical Logs (Text), Sensor (Time-series)	Advancing failure prediction for maintenance	[133–136]
GenAI-empowered Recycling & Remanufacturing	Collaborative Disassembly	Robotic Trajectories, Disassembly Instructions (Text)	Optimizing disassembly with human-robot collaboration	[137,138]
	Remanufacturing	Production Records (Text), Process Adjustments (Time-series)	Improving processes through adaptive strategies	[139]

these models to be reliable, the data they generate must accurately reflect real-world behavior, particularly in critical applications like industrial systems [140]. Interpretability, however, goes beyond accuracy—it requires that the generated data can be understood by domain experts, enabling informed decision-making. Unfortunately, many modern GenAI models lack intrinsic interpretability, functioning as black boxes, which makes explaining their outputs difficult [141]. To address these challenges, techniques such as probing [142], local interpretable model-agnostic explanations (LIME), and Shapley Additive explanations (SHAP) can improve transparency. For instance, identifying which sensor data influenced fault detection in a manufacturing system. SHAP, widely used in applications like fault diagnosis, assigns importance values to each input feature, providing a clear breakdown of their contributions to the output, such as highlighting critical variables in semiconductor yield prediction. Additionally, developing hybrid systems that combine complex deep learning models with more interpretable statistical models allows for fallback options when critical decisions require explanation. While these methods are not foolproof, they provide practical approaches to balance model performance with necessary transparency [143].

Hallucinations. Hallucination in GenAI refers to the phenomenon where models generate content that is not factually accurate, consistent, or grounded in the information provided during training or input context [144]. Hallucinations can arise both from inconsistencies within the context of the current conversation and from gaps or inaccuracies in the training data, making the generated information unreliable. In industrial settings, challenges include managing inconsistent data, preventing hallucinations in critical tasks like defect detection, and integrating with legacy systems, which can impede its accuracy and reliability. Cybersecurity risks and workforce skill gaps further complicate AI deployment, requiring robust validation and upskilling efforts. Mitigating hallucination involves ensuring that the model outputs verifiable and grounded responses, as well as having the ability to admit when it lacks sufficient information to provide a correct answer. While methods like RAG, output filtering, and importing external tools are commonly used to reduce hallucinations and provide up-to-date information for GenAI models, it can still produce inaccurate or even harmful content, which may negatively impact the final result [145].

4.1.2. Domain-specific requirements

Data-intensive systems, including GenAI-integrated systems, heavily depend on high-quality data to function effectively. Manufacturing processes generate data from a multitude of sources, including sensors, machines, human operators, and supply chain systems, which are often unstructured, noisy, and heterogeneous. Inaccurate, incomplete, or biased data can lead to erroneous outputs, affecting decision-making and operational efficiency [146]. For instance, discrepancies in data formats from different sensors and machines can impede data integration and analysis. This heterogeneity makes it difficult to create unified models that can effectively process and analyze all relevant data. Developing GenAI models that can handle diverse data types and sources is critical for capturing the full scope of manufacturing operations and providing valuable insights.

Implementing GenAI techniques in smart manufacturing requires a deep understanding of domain-specific knowledge and access to high-quality data. However, manufacturing data and domain-specific knowledge are often proprietary, sensitive, and locked within organizations. Companies are reluctant to share data due to competitive concerns or regulatory constraints, limiting the availability of datasets needed to train GenAI models, as well as downstream applications. Genetic models without embedded domain-specific knowledge or external tools may also fail to capture the intricacies of manufacturing processes. This limitation can result in models that misinterpret or oversimplify complex industrial scenarios, leading to suboptimal or even erroneous outcomes. Enhancing contextual understanding requires integrating domain-specific knowledge and developing models capable of reasoning about manufacturing-specific contexts.

4.1.3. Deployment & integration issues

Manufacturing operations require rapid responses from GenAI systems to ensure both efficiency and safety. However, integrating GenAI technologies into existing production environments is not straightforward. Many assume that improvements in GenAI automatically lead to manufacturing gains, yet the benefits depend heavily on how well these advanced models work with established systems. Legacy control systems and data infrastructure often require significant modifications to interface effectively with new AI models. For example, even if a GenAI system delivers low-latency outputs for tasks like welding, assembling, or picking, its impact is limited if it cannot exchange data smoothly with pre-existing production equipment.

Open vs. Closed models trade-offs. Open models, while prized for their transparency and adaptability, present significant risks. Their accessible weights and architecture allow anyone to modify and deploy them, which can lead to misuse (e.g., generating harmful content, disinformation, and deepfakes). These models may also inadvertently leak sensitive training data or reveal biases embedded in large-scale datasets, complicating efforts to ensure fairness and accountability. In addition, the reproducibility of open models can be undermined by incomplete documentation, differences in hardware, or variations in training processes, which may lead to inconsistent results. Closed models, while offering a controlled and consistent performance environment, are not free of challenges. Their proprietary nature makes it difficult for independent auditors to assess fairness, safety, or alignment with ethical standards, hiding potential vulnerabilities or model drift that may adversely affect users. Both model types also face issues related to scaling, energy consumption, and the lack of robust regulatory frameworks to govern deployment. Mitigation strategies include establishing stringent ethical guidelines, improving documentation and transparency, adopting regular security and bias audits, and fostering collaborative research initiatives to develop standardized best practices.

Integration with existing systems. Existing systems often operate on outdated architectures and data formats, which were not designed to support the dynamic requirements of advanced applications. Integrating these systems presents significant technical challenges due to differences in protocols, data formats, and system architectures, potentially leading to operational disruptions [147]. These incompatibilities lead to difficulties with data translation, slow processing speeds, and an increased risk of operational disruptions. Upgrading or replacing outdated infrastructure requires substantial investment and careful planning to avoid interrupting production [148]. Moreover, such systems may lack the robust security measures needed in today's interconnected environments, thereby exposing potential vulnerabilities. To address these issues, organizations have explored several mitigation strategies. One effective approach involves implementing middleware that serves as an intermediary to convert data formats and reconcile protocol differences between older and newer systems. Another strategy is to incrementally adopt service-oriented architectures (SOA), which break down legacy functionalities into modular services that can be gradual, reducing the risk of widespread system failure.

As manufacturing operations evolve, AI systems must scale to handle increased data volumes and growing process complexity. Ensuring that GenAI models continue to operate accurately over time requires regular updates and careful integration with both new technologies and existing production protocols. This continuous process of updating and integrating AI models into established systems presents an ongoing technical challenge that goes beyond the mere adoption of advanced GenAI technologies.

4.2. Industry 5.0 considerations

Industry 5.0, introduced by the European Commission in 2021 [149], builds upon the digital connectivity and automation of Industry 4.0 by

emphasizing three core objectives: human-centric design, ecological sustainability, and operational resilience. Unlike Industry 4.0, which primarily focuses on automation and machine interconnectivity, Industry 5.0 integrates human insight alongside intelligent systems. Skilled workers collaborate with machines, and processes are assessed not only for efficiency but also for their impact on human well-being and environmental boundaries. Industry 5.0 aims to synergize human creativity and advanced technologies to create human-centric, sustainable, and resilient manufacturing systems [4]. While GenAI technologies empower smart manufacturing by enhancing efficiency, personalization, and innovation, it also introduces several challenges. This section examines the key challenges in AIGC-empowered smart manufacturing within the domains of human-centricity, sustainability, and resiliency (Table 7).

4.2.1. Human-centricity

The successful deployment of GenAI in manufacturing hinges not only on technical performance but also on human factors. While Industry 5.0 envisions a synergistic relationship between humans and machines, achieving this balance requires overcoming obstacles related to trust, communication, and interoperability. Designing intuitive and user-friendly interfaces for complex AI systems is critical yet challenging. Poorly designed interfaces can lead to errors, decreased productivity, and user frustration. Balancing the complexity of advanced AI functionalities with ease of use necessitates a deep understanding of user needs and behaviors, as well as iterative design processes to ensure accessibility and usability.

Ethical issues. Concerns about data privacy, surveillance, and algorithmic bias can undermine public trust if AI systems lack transparency and accountability [150]. Ethical issues arise when AI decisions are not explainable or when biases in algorithms lead to unfair or discriminatory outcomes. Ensuring that AI systems operate ethically requires the development of robust ethical frameworks and adherence to principles that prioritize human rights and societal well-being. Implementing encryption techniques for data at rest and in transit helps protect information from unauthorized access. Regular security audits and compliance reviews verify that data collection and processing follow established policies. Privacy-preserving technologies, such as differential privacy and secure multiparty computation, allow for the analysis of data while keeping sensitive details hidden. Finally, strict access controls and multi-factor authentication measures reduce the likelihood of data breaches and help maintain accountability in data handling [151].

A significant challenge in implementing GenAI technologies is addressing workforce skill gaps and resistance to technological change. Employees may lack the necessary training to interact effectively with advanced AI systems, leading to underutilization of technology and fears of job displacement. Resistance often stems from apprehensions about job security, lack of understanding of AI benefits, and discomfort with altering established workflows. Overcoming these challenges requires comprehensive change management strategies, including continuous education and reskilling programs to equip employees with the knowledge and skills needed to collaborate with advanced technologies.

4.2.2. Sustainability

Achieving sustainability requires addressing a range of interconnected challenges. A primary issue is the absence of standardized metrics and comprehensive evaluations [152], which hinders companies' ability to accurately assess the ecological impacts of manufacturing processes, particularly in the context of adopting GenAI technologies. AI systems are often highly resource-intensive, consuming substantial amounts of energy and computational power, which may exacerbate carbon emissions unless effectively optimized for energy efficiency [153]. This paradox presents a critical dilemma: technologies intended to promote sustainability may inadvertently contribute to environmental degradation if not properly managed. Moreover, the high costs associated with implementing these technologies constitute a significant barrier, particularly for small and medium-sized enterprises, limiting their capacity to adopt sustainable practices [154]. This financial constraint, in turn, poses a substantial impediment to achieving broader sustainability objectives.

4.2.3. Resilience

Safety, security and cyberattacks issues. In the manufacturing sector, the integration of GenAI technologies introduces both opportunities and challenges in enhancing resiliency [155]. The increased connectivity and reliance on AI systems, including GenAI-integrated systems in manufacturing may heighten vulnerability to cybersecurity threats and unforeseen disruptions [156,157]. Protecting sensitive data and ensuring system integrity require robust cybersecurity measures capable of adapting to evolving threats. Cyber-attacks can lead to significant operational disruptions, financial losses, and damage to reputation [158]. Additionally, while certain types of disruptions are predictable, unforeseen events such as pandemics or natural disasters can still severely impact manufacturing operations [159]. Although some of the GenAI techniques could help to defend cyberattacks like Denial-of-Service (DoS) in certain scenarios [160], developing GenAI models that can account for rare and unpredictable events is complex,

Table 7
Comparison of Key GenAI implementation approaches based on industry 5.0 principles and key considerations.

Industry 5.0 Principles	Key Factors in GenAI	Key Implementation Approaches		
		Manually Collected Data & Created Model	Fine-tuning Open Source Pre-trained Model	Using Providers' API
Human-centricity	Data Privacy	High, data remains in-house, with minimal external exposure	High, data remains in-house, with minimal external exposure	Low, data shared with external providers
	Transparency	High, full access to models and data	Moderate, fine-tuning data and open model offer some transparency	Low, proprietary models & data
	Skill Requirements	High, requires in-house expertise	High, need advanced skills to customize models	Low, minimal implementation skills
Sustainability	Environmental Impact	Low to Moderate, varies on model design	Low, fine-tuning use less resource	Extremely High, external training requires vast resource
	Costs for SMEs	High, significant initial investment required	Moderate, fine-tuning can reduce some cost	Low to Moderate, pay-as-you-go
	Resource Efficiency	Low to Moderate, varies based on the implementation	Moderate, pre-trained enhances efficiency	Moderate to High, depending on provider
Resilience	Integration Complexity	Moderate to High, task-specific, varies with integration methods	Moderate to High, task-specific, varies with integration methods	Low to Moderate, general APIs easily integrate with tasks
	Cybersecurity Risks	Moderate, data kept internal	Low, internal synthetic data mitigates risks	High, data transmission to external providers
	Flexibility	High, full control over data and parameters	Moderate, limited by selected pre-trained models	Low, customization depends on the provider's API flexibility

requiring extensive data and sophisticated algorithms. To mitigate cybersecurity risks in GenAI applications for manufacturing, it is crucial to implement advanced intrusion detection and prevention systems specifically designed for these environments. Regular cybersecurity audits and vulnerability assessments should be conducted to proactively identify and address potential security gaps. Additionally, stringent data encryption and access control mechanisms are essential for safeguarding sensitive manufacturing data generated and processed by GenAI models [161].

Lack of global consensus on data governance. Different national priorities have led to fragmented regulations (e.g., AI Risk Management Framework [162] by NIST, EU's AI Act [163]), and international organizations like OECD [164], UNESCO [165] have separately developed principles emphasizing transparency, accountability, human rights, and safety. The diversity of these approaches complicates international cooperation, making it challenging to maintain consistent data quality and governance practices across global operations. Coordinated international frameworks are needed to address these challenges, promoting data sharing and quality while respecting privacy and security concerns.

Lack of real-world evaluations. In the manufacturing industry, the absence of real-world evaluations for GenAI presents significant challenges, particularly in complex, safety-critical, and regulated environments. Key challenges include the scarcity of diverse, high-quality datasets due to the proprietary and sensitive nature of manufacturing data, which hinders the creation of standardized benchmarks. The intricate nature of manufacturing processes complicates the definition of clear evaluation metrics, while the high stakes—such as potential safety hazards, costly downtime, or regulatory violations—make live testing risky and impractical. Additionally, the variability across manufacturing settings limits the scalability of evaluations. Mitigation methods to address these issues include leveraging synthetic data generation to create privacy-preserving, simulated manufacturing scenarios for risk-free testing. High-fidelity simulations of production processes provide a controlled environment to evaluate GenAI models without disrupting operations. Evaluation construction methods can further enhance this by developing task-specific evaluations (e.g., for predictive maintenance or quality control) and establishing tailored metrics like production efficiency or defect rates. Hybrid approaches, blending limited real-world data with synthetic datasets, and collaborative benchmarking initiatives among manufacturers, researchers, and AI developers can also foster robust, scalable, and standardized evaluation frameworks, ensuring GenAI reliability despite the lack of extensive real-world data.

In addition to these integration challenges, deploying AI technologies raises critical safety and security concerns. AI systems must be designed to prevent unauthorized access, protect sensitive data, and work without introducing vulnerabilities into the current manufacturing framework. The need to align these safety protocols with existing systems further complicates integration. Furthermore, the risk of adversarial attacks requires ongoing monitoring and advanced security strategies to secure both physical assets and personnel.

4.3. Research limitations

Despite our systematic approach, several limitations persist. First, the integration of GenAI in manufacturing is still in its early stages. As a result, comprehensive real-world case studies and empirical evaluations across diverse industrial settings are scarce. Although the framework for human-in-the-loop research curation is promising, our study currently lacks the conditions to conduct a full-scale empirical examination. While we have integrated real-world examples (e.g., Under Armour's example in Section 3.1) to demonstrate feasibility, these examples do not fully address all aspects of implementation, and may lack real-world pilot metrics due to limited availability of publicly accessible documentation. Second, even with extensive efforts, our coverage of the rapidly emerging GenAI developments is not exhaustive. Third, although we supplement academic search engines with industry white papers and blog posts, many critical insights remain proprietary. These limitations underscore the need for broader collaboration and suggest that future work should address these gaps through longitudinal studies and expanded data sources.

5. Prospects and research directions

The development of GenAI in manufacturing requires an integrated and comprehensive approach that combines advanced technology, human collaboration, and the use of high-quality data solutions. As illustrated in Fig. 6, this chapter highlights the importance of establishing a robust infrastructure, fostering effective human-GenAI collaboration, and developing various GenAI agents to address different manufacturing needs. It emphasizes the need to optimize model capabilities, uphold responsible AI practices, and leverage quality datasets to improve the overall performance of GenAI systems. The aim is to unlock the full potential of GenAI, enhancing efficiency, promoting innovation, and ensuring the manufacturing industry's adaptability and resilience in a continuously evolving environment.

5.1. Infrastructure

5.1.1. Responsible GenAI

Responsible GenAI is necessary to ensure that AI systems remain ethical, fair, and aligned with human values as they grow in power and autonomy. It encompasses the principles, practices, and frameworks aimed at ensuring that GenAI systems are developed, deployed, and managed in a manner that aligns with ethical, transparent, and sustainable standards [166]. It focuses on minimizing biases, promoting accountability, and ensuring fairness in AI-driven processes across industries. In the context of smart manufacturing, Responsible GenAI emphasizes secure, compliant handling of sensitive data to ensure generated content align with regulatory standards and organizational ethics while promoting fair, unbiased algorithms to mitigate societal biases.

Explainable GenAI. Explainable GenAI refers to GenAI systems that provide clear, understandable insights into how they produce outputs and make decisions, addressing the "black box" nature of traditional AI models [167]. It emphasizes transparency, user-friendly interpretability, traceability, and the detection of biases to ensure that users across industries, including smart manufacturing, can trust and comprehend GenAI-driven outcomes [168]. Several research advancements could improve explainability in GenAI, including Physics-informed Neural Networks, Liquid Neural Networks, and Untrained Neural Networks, each contributing uniquely: embedding physical laws, adapting in real-time to dynamic data, and utilizing inherent neural structures without pre-training, respectively. By offering interactive explanations and making decision processes accessible, explainable GenAI fosters greater confidence in GenAI recommendations, enabling human operators to work more effectively with AI in tasks such as predictive maintenance, process optimization, and automation. Further research in Explainable GenAI could focus on developing intuitive methods for explaining complex AI models, improving real-time interpretability for non-experts, creating interactive query interfaces, ensuring transparency in GenAI-driven automation, addressing fairness and bias, balancing model complexity with explainability, and integrating explainability seamlessly into industrial workflows.

Transparent and Democratized GenAI. Recent trends indicate a clear shift toward transparent and democratized GenAI, emphasizing open sharing of development processes and broad accessibility to GenAI tools. Transparency ensures that AI systems' data sources, algorithmic decisions, and deployment are visible and accountable, fostering trust and ethical use [169]. Democratization extends AI capabilities to a wider range of users, allowing employees with diverse expertise to automate tasks, generate insights, and drive innovation through intuitive interfaces [170]. Although a performance lag exists between open

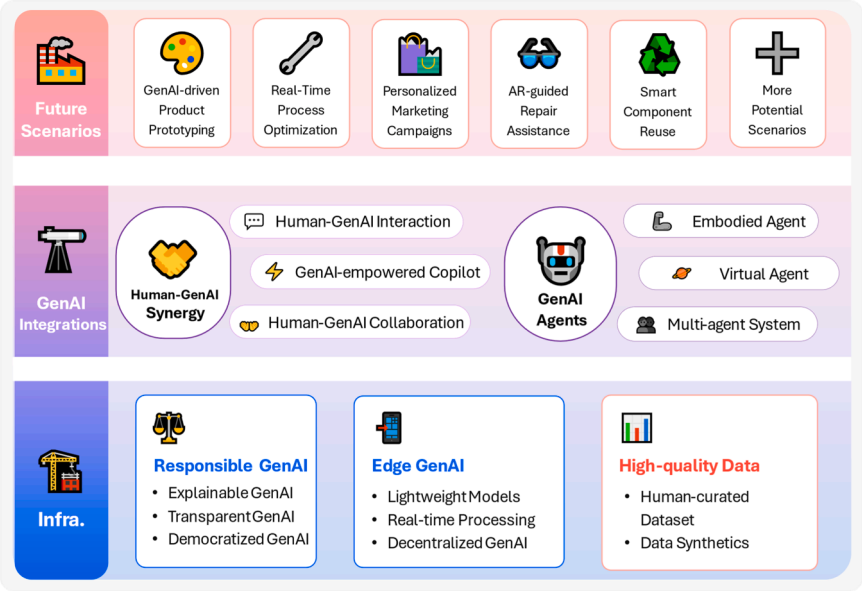


Fig. 6. Prospects, research directions and future scenarios landscape.

and closed models, new open models are closing the gap by reaching high levels of performance with lower training compute requirements. At the same time, the growth of community platforms and shared resources has boosted innovation, while major companies continue to invest in closed models to offer high-performance, commercially driven solutions. Future research could focus on ethical frameworks, secure handling of industrial data, enhancing GenAI accessibility for diverse workforce skill levels, fostering cross-functional collaboration, and the broader societal implications of GenAI-driven automation in smart manufacturing.

5.1.2. *Edge GenAI*

In contrast to the trend of scaling up, there is a significant movement towards developing small generative models suitable for edge deployments and resource-constrained environments. An all-encompassing large-scale model pales in comparison to specialized small models woven seamlessly into every field. Lightweight models (e.g., GPT-4o mini [171], Phi [172]) offer competitive performance, serving as a practical alternative to their larger counterparts while being more accessible for fine-tuning and incurring lower operational costs [173]. Small generative models are particularly valuable in scenarios where efficiency and latency are critical, such as on-device applications and real-time processing. Their development expands the reach of GenAI technologies into mobile and embedded systems, enabling intelligent functionalities in edge devices. Moreover, they support the growth of decentralized AI applications, enhancing privacy and reducing reliance on cloud-based services.

By processing real-time data on local devices such as IoT sensors, smart machinery, and industrial equipment, Edge GenAI enables real-time decision-making and faster responses, critical for optimizing production lines, predictive maintenance, and quality control in manufacturing [174]. This decentralized approach reduces latency, enhances performance, and ensures greater data privacy by minimizing reliance on cloud systems [175]. Recent advancements in small generative models have expanded the potential of Edge GenAI in smart manufacturing. These lightweight models offer efficient, real-time processing on resource-constrained devices, enabling intelligent functions like predictive maintenance and process optimization in manufacturing environments. Their development reduces latency, operational costs, and reliance on cloud infrastructure while enhancing privacy and scalability, making Edge GenAI more practical for decentralized AI applications across the whole manufacturing process. Integrating existing technologies, such as blockchain with Edge GenAI strengthens data privacy, fosters trust and boost resource efficiency by facilitating secure data sharing, federated learning, and auditable decision-making across decentralized devices.

5.1.3. *Human & synthetic datasets*

High-quality training data plays a critical role in the performance, accuracy, and quality of responses from custom models, especially for data-driven. The quality of human datasets is paramount, as high-quality, diverse, and well-labeled data enables GenAI to generalize effectively across tasks, reducing bias and enhancing the accuracy and creativity of its generative capabilities. In industry, there are some high-quality human-curated datasets, such as AssemblyHands [176], HA-ViD [177] for industrial assembly tasks. Considering the scarcity of domain-specific data in publicly available datasets, developing high-quality human-generated datasets is a promising long-term direction.

Although human-curated datasets are typically high-quality, acquiring robust datasets can be prohibitively expensive and challenging to access. To address these challenges, synthetic data offers flexibility, cost-efficiency, and scalability, enabling specialized models to achieve relatively high performance even in data-scarce environments [178]. In the realm of smart manufacturing, traditional data synthetic methods are mainly based on physical simulations and data-driven numerical modeling [179]. Recent advancements in generative models have significantly enhanced the field of synthetic data generation. In manufacturing, synthetic data generation using generative models opens new possibilities for creating high-quality and contextually rich datasets encompassing multiple modalities (e.g., time-series data from sensors, video, and image with annotations) adopted for imbalanced data learning, safeguarding sensitive information, or improving data quality by denoising data across the whole PLM. In some cases, state-of-the-art generative models even surpass the performance of crowd workers [180,181]. Promising research directions are emerging in enhancing manufacturing systems' ability including synthetic data, GenAI-driven data augmentation for further training [182], generate simulation, or even generate simulation environment [183].

5.2. Human-GenAI synergy

5.2.1. Human-GenAI interaction

Human-centered GenAI focuses on aligning AI systems with human values, understanding human intentions, and augmenting human abilities, which highlight ethical standards and enhance human productivity, ensuring GenAI remains intuitive and beneficial to users. Emphasizing human-centered design principles enhances the usability and acceptance of GenAI systems. In manufacturing, intuitive interfaces and natural language processing enable operators to interact seamlessly with GenAI systems, improving decision-making and operational efficiency. Early explorative research mainly focused on enhancing the human-machine interface [184,185]. By facilitating transparent and user-friendly interactions, manufacturers can bridge the gap between complex GenAI algorithms and human operators. Empowered by the multimodal perception and understanding capabilities of state-of-the-art GenAI models, developing more universal and human-friendly interfaces represents a promising direction for researchers to pursue. Besides, multimodal capabilities of state-of-the-art models like VLMs can significantly enhance human-machine interfaces in manufacturing environments. These models enable visual instruction interpretation where operators can show images of production issues to GenAI systems which can interpret them alongside verbal descriptions, creating more intuitive communication channels. They can also facilitate context-aware assistance by understanding the visual manufacturing environment and providing guidance based on real-time visual input, allowing for more adaptive support systems.

5.2.2. GenAI-empowered copilot

The concept of a GenAI-empowered copilot involves AI systems assisting human operators by providing real-time insights, suggestions, and automation of routine tasks. Such augmentation enhances efficiency and reduces the cognitive load on workers. Recent advancement in such copilots includes software engineering (e.g., GitHub Copilot [186]), with capabilities for automating tasks such as code generation, testing, debugging, and documentation. In the smart manufacturing domain, early exploration has been using fine-tuned LLM for troubleshooting [187]. By serving as copilots, GenAI systems help workers navigate complex processes and make informed decisions promptly, leading to improved performance and reduced errors.

5.2.3. Human-GenAI collaboration

Human-GenAI collaboration represents an advanced partnership where GenAI systems not only assist in manufacturing processes but also actively perceive and respond to human physiological and psychological states. In smart manufacturing, this means integrating sensors and AI algorithms that monitor indicators like fatigue, stress levels, and cognitive load among workers [188]. By analyzing this data, GenAI systems can adapt workflows in real-time—redistributing tasks, adjusting machine operations, or providing timely alerts—to enhance both productivity and worker well-being. This approach transcends traditional human-machine interactions, moving beyond mere operation or supervision of equipment. Machines become empathetic collaborators who understand human needs at a deeper level. For instance, if a worker exhibits signs of stress, the system might slow down the production line slightly or provide additional support resources. Such responsiveness not only improves efficiency but also fosters a safer and more supportive work environment. Early explorations in smart manufacturing focused on basic human-robot collaboration and simulation modeling of logistic systems. By elevating this collaboration to include the sensing of human physiological and psychological changes, manufacturers can achieve higher levels of synergy between humans and machines. VLMs and VLAs enhance collaboration beyond physiological sensing by introducing task-aware visual assistance that can observe human workers performing complex tasks and provide just-in-time visual guidance precisely when needed [189]. These models enable sophisticated gesture and intent recognition to interpret human movements and understand worker intentions without requiring explicit verbal commands, creating more natural interaction patterns on the factory floor. Furthermore, VLM-powered augmented reality collaboration interfaces can overlay visual instructions precisely aligned with the physical work environment, combining the strengths of human dexterity and adaptability with AI's precision and knowledge repository, resulting in more efficient and error-resistant manufacturing processes. This advancement paves the way for more flexible production lines, optimized workflows, and enhanced safety protocols, ultimately leading to a more resilient and adaptive manufacturing ecosystem.

5.3. GenAI agent

GenAI agents are intelligent systems built on GenAI models that enable them to autonomously interact with their environment, perceive real-time inputs, and execute tasks without direct human intervention (Fig. 7). In general, these agents integrate perception, advanced planning, memory management, tool use to operate autonomously. The perception module acts as the sensory interface: text inputs are processed by LLMs, images by vision-language models (VLMs) or computer vision techniques, and audio by speech-to-text systems. This robust sensory integration allows the agents to gather and interpret multimodal data, forming the basis for informed real-time actions. The planning module employs techniques such as Chain-of Thought (CoT) [61], Tree of Thoughts (ToT) [62], and ReAct [190] to break down complex tasks and generate step-by-step reasoning, while Reflexion [191] uses past actions to inform future strategies. In addition, methods like Chain-of-Hindsight (CoH) [192] and Algorithm Distillation (AD) [193] further optimize decision-making through RL approaches. Memory management combines short-term in-context learning with long-term retrieval from external vector stores using approximate nearest neighbor methods such as FAISS [194], ensuring that both recent and historical data are readily. Complementing the core architecture, the tool use component expands the functionality of GenAI Agents by interfacing with external systems and resources. Frameworks such as MRKL [195] direct queries to specialized expert modules, while TALM [196] and Toolformer [197] enable the autonomous invocation of APIs. Benchmarks like API-Bank [198] and implementations such as HuggingGPT [199] assess and guide this tool augmentation.

Distributed Artificial Intelligence (DAI) agent is an autonomous software process operating on a single node within a networked system that independently senses its local environment, maintains internal knowledge, performs reasoning, and communicates with peer agents to contribute to distributed problem-solving [200]. Reinforcement Learning (RL)-based agents, by contrast, are designed to optimize sequential decision-making through learned policies. GenAI agents are distinguished by their focus on generative reasoning and interactive capabilities (Table 8). Integrating GenAI agents into manufacturing environments presents several challenges, including ensuring reliable long-term memory, enhancing long-term planning capabilities, and improving reasoning skills. Additionally, the seamless integration of GenAI agents with existing manufacturing systems requires robust infrastructure and interoperability standards. Future research could focus on developing algorithms or frameworks that enhance GenAI agents' memory and reasoning, creating standardized protocols (e.g. Claude's Model Context Protocol) for agent integration, and exploring hybrid models that combine the strengths of GenAI, MAS, and RL agents. Moreover, addressing ethical considerations and ensuring transparency in GenAI agent decision-making processes will be essential for fostering trust and acceptance among human operators.

5.3.1. Embodied agent

An embodied GenAI agent is an intelligent system that combines GenAI with a physical presence, allowing it to autonomously perceive, interact, and respond to its environment in real time. Embodied GenAI

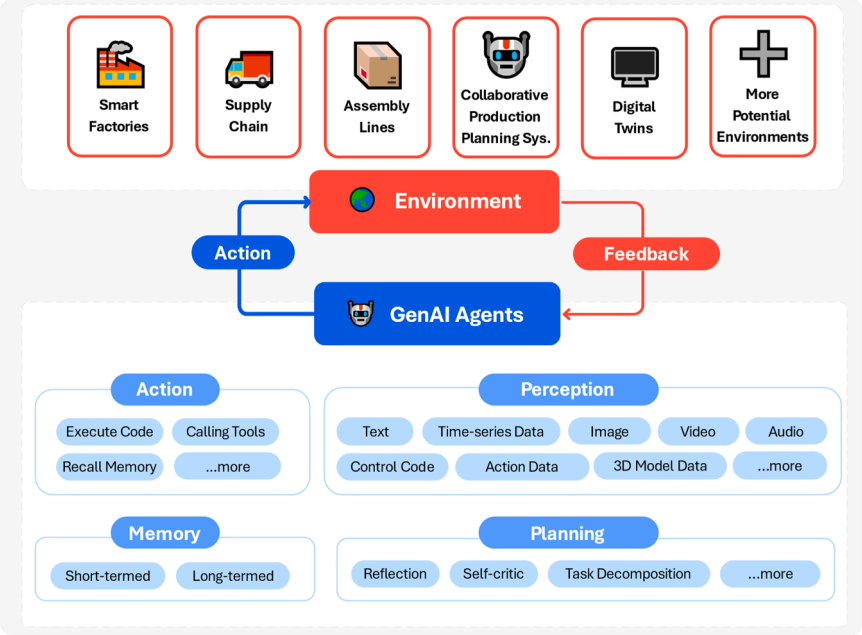


Fig. 7. A basic GenAI agent framework.

Table 8
Comparison between GenAI agents, DAI agents, and RL agents.

Aspects	GenAI Agents	DAI Agents	RL Agents
Core Focus	Real-time adaptation, reasoning, and content generation	Collaboration and decentralized problem-solving	Sequential decision-making and skill optimization
Capabilities	Leverages GenAI models for human-like decisions and content generation	Features autonomy and inter-agent communication	Learns optimal behaviors through trial and error
Environment	Operates in both virtual and physical settings	Functions in decentralized, dynamic environments	Operates within simulated or controlled environments
Learning Approach	Utilizes pre-trained and domain-specific knowledge from GenAI models	Adapts through collaboration and communication	Trains through reinforcement learning and feedback
Applications	Enhances decision-making, content creation, and human interfacing	Optimizes resource allocation and task execution	Excels in robotics and high-dimensional task management

agents, equipped with physical capabilities, have the potential to revolutionize smart manufacturing by performing tasks that require interaction with the physical environment. With the enhancement of state-of-the-art LLMs and VLMs, embodied agents have shown great promise in handling complex, repetitive, or hazardous tasks (e.g., TidyBot [201], ProAgent [202]). In the future, these agents, in the form of autonomous robots or machines, can finally execute tasks with high precision, such as assembling components, packaging, or inspecting products for quality control. With the ability and adapt in real time, embodied GenAI agents can improve productivity, reduce errors, and increase workplace safety. Their physical presence allows them to seamlessly integrate with both human workers and automated systems, enabling more flexible and adaptive manufacturing processes.

5.3.2. *Virtual agent*

In contrast, virtual GenAI agents exist solely in digital environments. Several early explorations in virtual agent encompass various domains, including industrial automation (e.g., TaskMatrix.ai [203]), software engineering (e.g., DevinAI [204]), and scientific discovery (e.g., the AI Scientist [205]). Applications including ChemCrow [206] for chemistry-related tasks and the Generative Agents Simulation [207] that models human-like interactions illustrate how these agents combine structured reasoning with practical tool use. Nonetheless, challenges remain: limited context lengths, potential model biases, and the difficulty of processing noisy or incomplete sensory data limit the agents' ability to perform extended tasks or manage unexpected changes in real time. Virtual GenAI agents hold significant potential in smart manufacturing by enhancing decision-making, optimizing workflows, and automating digital processes. They can monitor production systems in real time, analyzing vast amounts of data from sensors and machinery to predict maintenance needs, identify inefficiencies, and suggest improvements. They also power virtual simulations for testing new manufacturing processes or products without disrupting actual operations. Additionally, virtual GenAI agents handle tasks like inventory management, demand forecasting, and supply chain optimization, ensuring smoother and more efficient operations in smart factories through their adaptive learning capabilities.

5.3.3. *Multi-agent system*

Multi-agent systems (MAS) in manufacturing consist of multiple autonomous agents working collaboratively to optimize production processes. GenAI-based multi-agent systems have adapted to domains like software development (e.g., ChatDev [208]) and social simulation (e.g., Generative Agent [209], AgentSims [210]). In manufacturing, MAS exploration has been made in a conceptual framework [211], dynamic scheduling [212], process control [213], and autonomous manufacturing [214]. With advancements in GenAI technologies, each agent within a MAS can independently monitor, make decisions, and control specific tasks like machinery operation, inventory management, or quality control. Coordinated actions among agents improve flexibility, efficiency, and responsiveness in smart manufacturing environments, handling complex tasks such as production scheduling, resource allocation, and real-time process adjustments. This coordination makes the system more adaptable and scalable, enhancing overall manufacturing performance.

6. Concluding remarks

This study has elucidated the transformative potential of AIGC in the realm of smart manufacturing, highlighting its multifaceted applications across all phases of PLM. Through a holistic exploration of AIGC infrastructure, customization methods, and typical GenAI applications, the research underscores the significant advancements and efficiencies that GenAI can introduce to manufacturing ecosystems. While AIGC promises significant advancements in efficiency and innovation within manufacturing ecosystems, its practical integration is not without critical challenges that must be addressed.

A primary hurdle lies in model interpretability and reliability; the inherent "black box" nature of many sophisticated GenAI models impedes trust and adoption, particularly in safety-critical manufacturing applications where a clear understanding of the decision-making process is paramount for ensuring operational reliability (e.g., [140,141]). Compounding this is the challenge of data heterogeneity and domain specificity. Manufacturing environments generate vast quantities of data that are often unstructured, noisy, and sourced from diverse, frequently proprietary systems. This inherent heterogeneity, coupled with the necessity for deep domain-specific knowledge to interpret such data accurately, complicates the development of robust and broadly generalizable GenAI models [146]. Furthermore, the practical deployment of GenAI is hampered by difficulties in integration with existing legacy systems and the imperative to maintain robust cybersecurity. Effectively interfacing advanced GenAI capabilities with established manufacturing infrastructure, while simultaneously safeguarding against new and evolving cyber threat vectors, presents a substantial barrier to widespread and secure adoption [147,156,157].

To navigate these complexities and fully unlock AIGC's potential, future research must urgently prioritize several distinct, actionable initiatives. There is a pressing need to develop advanced Explainable GenAI methods, focusing on techniques such as hybrid models or physics-informed neural networks that can provide clear, understandable insights into GenAI decision-making, thereby enhancing transparency and user trust in critical manufacturing operations [167,168]. Equally vital is the engineering of lightweight and efficient GenAI models tailored for edge deployment. This involves investigating model compression, quantization, and specialized architectures to enable real-time AIGC applications on resource-constrained edge devices, which would reduce latency and bolster data privacy directly on the factory floor [174,175]. Progress in creating robust GenAI agents with enhanced long-term planning capabilities and sophisticated multimodal reasoning will also be pivotal; this requires advancing agent architectures (e.g., [190]) to improve long-term memory, handle complex planning tasks, and effectively integrate diverse sensory inputs—visual, textual, and sensor data—for autonomous task execution within dynamic manufacturing environments. Finally, establishing standardized frameworks for the secure generation and sharing of synthetic data is crucial. Developing robust protocols and platforms for creating high-quality, privacy-preserving synthetic datasets specifically designed for manufacturing applications will foster collaboration and accelerate model development, particularly in domains where real-world data is scarce or highly sensitive [178,182].

Addressing these specific challenges and pursuing these targeted research directions will be instrumental. The continued evolution of GenAI, guided by principles of responsible innovation and fostered through interdisciplinary collaboration, is poised to unlock new frontiers in efficiency, sustainability, and resilience, thereby shaping a more adaptable and intelligent industrial landscape for the future.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiewu Leng: Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Keyou Zheng:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Rongjie Li:** Writing – review & editing, Investigation. **Chong Chen:** Writing – review & editing, Conceptualization. **Baicun Wang:** Writing – review & editing. **Qiang Liu:** Writing – review & editing, Supervision, Funding acquisition. **Xin Chen:** Supervision, Funding acquisition. **Weiming Shen:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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